A Deep Learning Benchmark on Fashion-MNIST: From Multilayer Perceptrons to Advanced CNN Architectures and Interpretability

Ambeer Shravan Kumar
Roll No: 160123737168
Class: IT-3
Information Technology Department

***Abstract*—This paper presents an extensive empirical evaluation of deep learning methodologies applied to the Fashion-MNIST dataset. We investigate the performance of Multilayer Perceptrons (MLPs) under varying hyperparameters, including learning rates and optimization algorithms such as Adam, RMSProp, and Nesterov Momentum. Furthermore, we transition to convolutional neural network (CNN) architectures, evaluating classical models like LeNet-5 and more complex structures such as ZF-Net. Beyond classification accuracy, we employ visualization techniques, including feature map extraction and Guided Backpropagation, to interpret hierarchical feature learning. Finally, we demonstrate the utility of Denoising Autoencoders for spatial feature compression and reconstruction. Our results confirm that while MLPs achieve competitive benchmarks (87%), CNN architectures such as ZF-Net provide superior spatial awareness, reaching accuracies of over 90.7%.**

***Index Terms*—Fashion-MNIST, Deep Learning, MLP, CNN, ZF-Net, Guided Backpropagation, Autoencoders.**

I. Introduction

The classification of clothing items is a fundamental challenge in computer vision with significant applications in e-commerce and retail automation. While the original MNIST digit dataset served as the primary benchmark for decades, its simplicity often led to over-optimized models that failed on more complex data. Fashion-MNIST was introduced as a direct drop-in replacement, consisting of 70,000 grayscale images of fashion products across 10 categories.

This study aims to provide a systematic analysis of this dataset using a spectrum of neural architectures. We begin with foundational Multilayer Perceptrons (MLPs) to establish baseline performance and study the impact of optimization strategies. We then move into the convolutional domain, where spatial hierarchies are preserved. The paper further explores model interpretability to visualize how these networks "see" clothing items and concludes with generative reconstruction using autoencoders.

II. Dataset and Preprocessing

The Fashion-MNIST dataset contains 60,000 training samples and 10,000 test samples. Each sample is a 28×28 pixel grayscale image. The classes include: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, and Ankle boot.

A. Preprocessing Pipeline

To ensure efficient gradient descent, all pixel intensities were normalized from the range $[0, 255]$ to $[0, 1]$. For MLP architectures, the 28×28 images were flattened into 784-dimensional vectors. For CNN architectures, the images were reshaped to $(28, 28, 1)$ to maintain the spatial correlation of pixels.

III. Multilayer Perceptron (MLP) Analysis

We conducted multiple experiments to determine the sensitivity of MLPs to architectural and training hyperparameters.

A. The Impact of Learning Rate (η)

The learning rate is the most critical hyperparameter for convergence. Our experiments revealed:

- **Optimal Rate (0.001):** Training loss decreased smoothly, with validation accuracy reaching approximately 87.41%.
- **Too High (0.1):** The model failed to learn, with accuracy hovering around 10%.

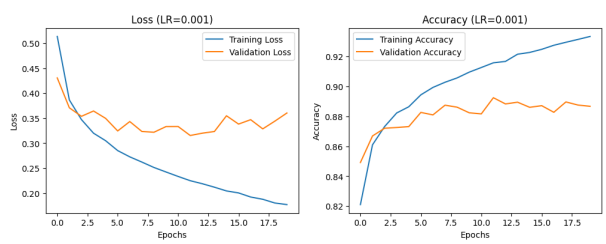

Fig. 1: Training vs Validation Loss and Accuracy for MLP (Learning Rate Analysis)

B. Architecture Complexity

We observed that a single hidden layer with 128 to 256 nodes provides sufficient capacity to separate the 10 fashion classes. While deeper architectures might provide marginal gains, they increase the risk of overfitting.

C. Optimizer Comparison

A comparison of different optimization strategies was performed using a baseline MLP ([128] nodes, 20 epochs). The results are summarized in Table I.

TABLE I: MLP Performance Across Different Optimizers

Optimizer Strategy	Test Loss	Test Accuracy
Standard SGD	0.4669	83.49%
SGD + Momentum	0.3565	87.47%
Nesterov Momentum	0.3684	86.85%
Adagrad	0.4189	85.29%
RMSProp	0.4236	86.40%
Adam	0.3542	87.28%

IV. Convolutional Neural Networks

Unlike MLPs, CNNs utilize convolutional filters to preserve the spatial hierarchy of clothing items.

A. Comparative Analysis of CNN Architectures

We evaluated several standard architectures to determine their efficacy on fashion imagery.

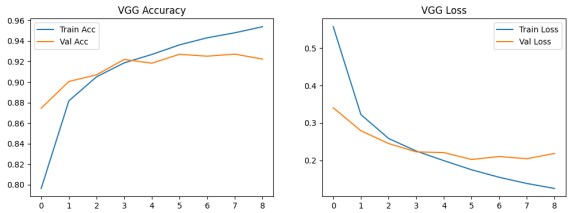

Fig. 2: VGG Model performance metrics on Fashion-MNIST

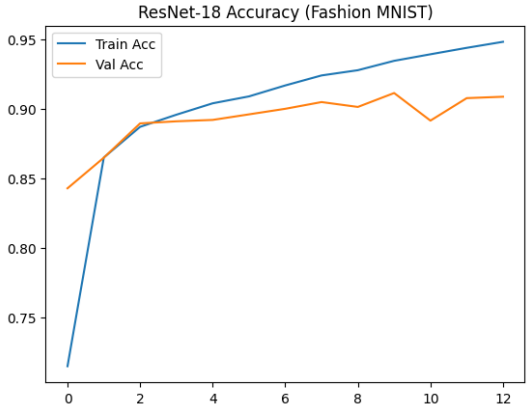

Fig. 3: ResNet-18 Accuracy and Loss convergence

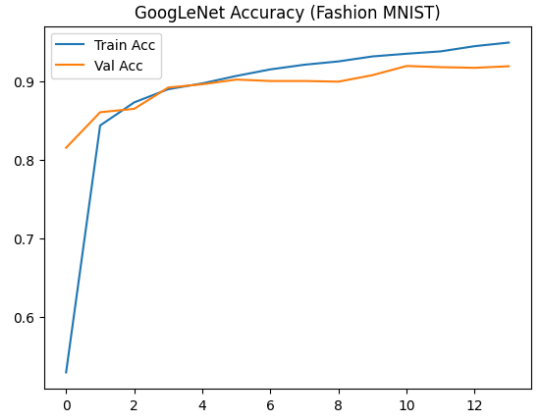

Fig. 4: GoogLeNet Inception-based performance evaluation

B. ZF-Net Performance

ZF-Net demonstrated superior feature extraction capability for this dataset, achieving a test accuracy of 90.70%.

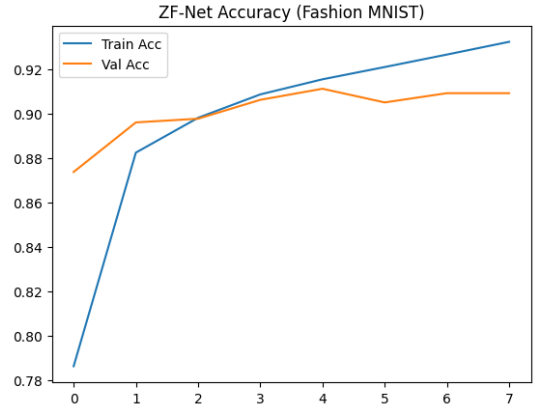

Fig. 5: ZF-Net Training and Validation Accuracy

V. Model Interpretability and Visualization

To understand model behavior, we visualized internal activations and used backpropagation-based methods.

A. Feature Map Visualization

Convolutional filters act as "specialized flashlights," highlighting edges and textures.

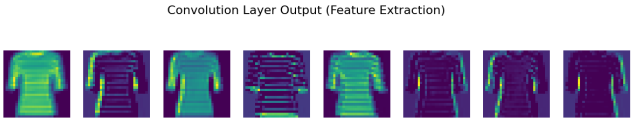

Fig. 6: Feature maps extracted from convolution layers showing learned spatial patterns

B. Guided Backpropagation

Guided Backpropagation was used to produce hyper-specific edge detection maps, highlighting the pixels the model deems most important for classification.

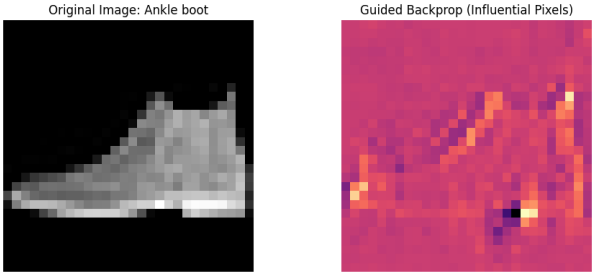

Fig. 7: Guided Backpropagation highlighting critical pixels for the 'Ankle Boot' class

VI. Denoising Autoencoders

We implemented a Denoising Autoencoder to learn compressed representations of clothing items and remove synthetic noise.

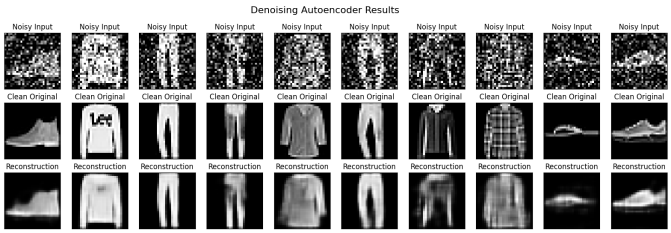

Fig. 8: Denoising autoencoder: Comparison between noisy and reconstructed outputs

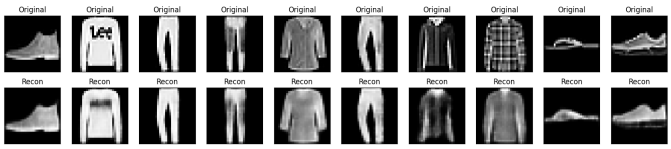

Fig. 9: Original vs reconstructed images showing feature compression quality

VII. Conclusion

This study highlights the transition from standard linear models to spatially-aware convolutional architectures. While optimized MLPs provide a strong baseline (87% accuracy), modern CNN architectures like ZF-Net push performance past the 90% threshold. Visualization tools and Autoencoders further provided deep insights into the network's internal representations.

References

1) Ambeer Shravan Kumar, "MLP Parameter and Optimizer Exploration on Fashion-MNIST," *Untitled29.ipynb*, 2026.
2) Ambeer Shravan Kumar, "Autoencoder Reconstruction and Denoising on Fashion-MNIST," *DL_week11(168).ipynb*, 2026.
3) Ambeer Shravan Kumar, "Guided Backpropagation for Model Interpretability," *DL_week10(168).ipynb*, 2026.
4) Ambeer Shravan Kumar, "Visualizing Convolutional Feature Maps," *DL_week9(168).ipynb*, 2026.
5) Ambeer Shravan Kumar, "ZF-Net Architecture Performance Analysis," *DL_week8(168).ipynb*, 2026.
6) Ambeer Shravan Kumar, "LeNet-5 Digit and Fashion Comparison," *DL_week7(168).ipynb*, 2026.
7) Ambeer Shravan Kumar, "Baseline CNN and Preprocessing Pipeline," *DL_week6(168).ipynb*, 2026.
8) Ambeer Shravan Kumar, "Regularization Techniques for Robust Image Classification," *DL_week5(168).ipynb*, 2026.